

I: We just have a note regarding the Covid-19 disruption. Obvious teaching has changed since we've all gone remote. And it's been disrupted. So, if you want to answer for any of the questions. In reference to your pre coated your normal teaching or to how teaching has changed. Either is fine. You can talk to one or both.

UCSB05: Understood.

I: Alright. So can we start with some background just on one, how your teaching relates to your current and past research. And also, much of the courses that you teach the students work with data.

UCSB05: Yeah, so, my current and past research focuses on [redacted field] is specifically around the area of human information behavior. And basically how people look for information how they seek it would they do with it, how they how they use the technology around it. I have a prior background to that in engineering, I was a [professional title redacted] for many years before I changed careers and became an academic and, and got my PhD [degree redacted]. So with that background that I have and now I'm really more interested in, issues of, you know, answering questions of social science, using computational methods that are pretty common in the [field redacted], which is where my background initially was. So that really informs a lot of what drives me to teach the classes that I teach here UCSB does not have an [redacted field] Department per se. So I am shared between the [redacted] department and the [redacted] department, which I love, I love having a foot in both those worlds. I have noticed that it's not a common thing on this campus. I am a lecturer so I'm not focused on research, currently, as much as I am on teaching.

So, the classes that I teach in the [redacted] department are [redacted], but in the [redacted] department I teach a mix of both classic undergrad courses like research methods and also I teach courses that really are very much about data driven and computationally data driven research so I teach social networks to undergrads and I teach. Also, a special topics class that I created for the department of [redacted department]. The both of those classes, especially the latter though, utilize or introduce the students to the notions of big data. And in social networks. I don't take it too far with the data, because it is about introducing them to theory and it actually uses a lot of [redacted field] stuff as well as you know application to come theory but the [redacted course], it's all about data, it's all about the use of data so about a third of that class is on a mixture of like an intro of what it all means the technology required, and the ethical and societal impacts that big data has. I'd say it's more than a third, I'd say it's almost half of the class is on that. I also do a module on data visualization. And I do a small module on coding and programming or are the very basic fundamentals of programming, given that the audience in that class is not are not computer science majors. But, you know, I want to introduce them to the notions of what it takes to program something with data. And some other some other minor things that I talk about but that's, that's a good summary of that.

I: I think we'll need to focus on the [discipline redacted] courses since it's social sciences, the focus for this study. So is the [data science course], do you use data sets in that course for doing the data, I imagine for the data visualization and the coding?

UCSB05: I do, and I introduced... I obtained a couple of data sets. We talked about also how one might obtain data sets as well so I talked them about some of the technicalities around. Finding data online using web scraper technology or using API technology to download data sets. We don't take it too far because again the audience is not really primed for to go into in too much technical depth. But I then provide them with like a small, or an example of what a big data set might look like and walk them through have them do exercises with analysis, and some basic statistical analysis of it, some visualizations. This has turned out to be more of a survey course like, here's a bit of this here's a bit of that without delving too deep. I would love to come up with another class that say, is all about data visualization and actually have them do a database of their projects with that, but that's for the future.

I: Yeah, yeah. the next question is about how the students obtain the data that they're working with, whether they collect are generated themselves, whether they're searching are identifying, pre-existing data, or whether you provide the data set and it sounds like you give them an overview of all of those things but then you end up giving them that that data set to work actually work with and in the class.

UCSB05: That is correct. And I would have liked to experiment with having them, collect the data themselves or like a [unintelligible] exercise. However, when I came up with this class, I was slated to teach it in the spring last year. And as you know, we went, because of COVID, we went online with it. And so, I lost the ability to. I was planning on the TA and myself doing, you know, a fair amount of hand holding with these non-technical students to kind of show them on a computer. Look, do this, do that and this is how you get it right. And that's really hard to do online. So, I had to change my plans and curtail that, which is why I decided to tell them about, you know, hey, theoretically, this is how you go about and get it but look, I have one for you. Here you go. Open up excel.

I: Yeah, we've been experiencing that with the carpentry workshops too. Yeah, I imagine for finding data sets, you know, there's all sorts of other scenarios that students might find themselves in and can't dig themselves out of

UCSB05: Yeah, yeah exactly and the variability I found with these students who are social science students, mostly [discipline redacted] majors, the variability is very big, in terms of how easily they use technology. Some of they'll jump right in, and they'll try this and that and others are just like what button do I press, ya know, they're a little more afraid of it.

I: Yeah, yeah, afraid of doing something irreversible.

UCSB05: Yeah, yeah that sense of that sense of experimentation isn't always there with undergrad students.

I: Yeah. Do you find it difficult to find those data sets that you use for that course, or did you have to do any kind of grooming of them to make them appropriate for the class?

UCSB05: Yeah, so, no it's not difficult to find data sets online so many sources. And, and, yes, eventually I did have to groom the data set, as you said, mostly just by taking out categories of the data like taking out fields, though, for example, though, the one assignment that I gave them was to analyze this data set of movies, there's this data set on movies on like their financial performance and, you know, how much profit they made and all of that. So it's, you know, there's a lot of categorical data and numerical data it's a nice mix of different types of data. The big data set that I acquired was, I mean most datasets that you can find out there are either CSV or, or JSON and CSV is so easy to open up with something like Excel it's pretty common place now. Yeah, and like I said I just had to like clean up some things like remove some uses a very big data set I removed some categories that I didn't think they needed to work with. but that was minimal.

I: Okay. And I should ask how large is this course, the [data science course]?

UCSB05: Yeah, both this one and the social networks course in [redacted course], are both upper division, and we get at minimum, 80 students, maybe up to 100, 110. So, yeah, average about 100 students.

I: So you have quite a few students that you're working with and do they then have like the actual lecture and then a lab, as well, with TAs or is it just lecture based?

UCSB05: it's just lecture based. There is no lab with this, it would have been fantastic. but no.

I: All right, so I think you've touched on this already a little bit but the next question is, you know, how do those students in that course, I'm thinking of the [data science course] specifically, how do they manipulate analyze and then interpret data in that course.

UCSB05: Um, so the exercise that I give them is...let me think, because I didn't know how many of them had access to Microsoft Excel. It's nice to assume that everyone can get that because actually I think they can get it for free on campus, if I'm not mistaken, but I ended up putting the data set on Google Sheets, sharing that with them. And it was a large data set, but it fit in Google Sheets so you know it's not like really big data. And what I had them do was do things like you know for categorical data, I taught them how to do frequency counts like you know how many times does this category appear versus that category and then like create a graph to represent it very basic data analysis type of tasks. Other numerical tasks, or you know, I'm sorry, categories that I mean, fields data types that were not categorical but were interval types or racial types, you know, that they could actually perform stats on. I taught them how to do, or I had them do you know find the mean, the median, the standard deviation, do a regression analysis between two variables, do a correlation between every pair of variables, things that they had learned in their...because they all have to take a beginners class in stats. You know

how much of that sticks in their head either? Not that much. At least they all remember what an average was that was something.

I: I can say from my intro stats I don't know how much I retained. So, it's not surprising.

UCSB05: But you know the [discipline redacted] department is actually very quantitative oriented. And a lot of the professor's there, you know their research is very quantitative, and they really push their students to know this stuff. But despite that predilection, a lot of the students needed a lot of refreshing. And that's fine, I spend half the lecture just reminding them what this stuff is. But then I give them the task to analyze the data that way and create graphs from it. In my short experience right, I've only taught this class once in the spring, and once in the summer and then in summer it was a lot shorter time and just wasn't quite the same thing. So, I don't have a big end to work with here but they did fairly well with it.

I: Okay. And so, were you able to do all of that in Google Sheets with them and, when you're demonstrating, do you show both in Excel and in Google sheets, are they close enough that people can figure it out?

UCSB05: No. A: it's close enough. B: what I was trying to avoid was giving them too much information and getting them confused because I know from past experience when I show them different tools, when my goal isn't to really teach them differences in the tools. My goal is to teach them the technique. And I showed them different tools. I get flooded with questions of "why does it work here and it doesn't work there?", and you know it's because they're different, you can't do the same thing exactly. So no, I stuck with one tool, thank you. With Google Sheets and the fact that Google Sheets is free and accessible and shareable and they could share with me what they were doing so I could look into what they were doing. It has a lot of advantages over Excel in that sense.

I: And did you expect the students to have prior knowledge of Google Sheets and whatever other tools they ended up using in that course? Or did you teach it all from scratch?

UCSB05: No. Well, I did not question their background on using Google Sheets, I believe, there's no study for this but anecdotally, and from personal experience, most of these undergrads by the time they're seniors, they've used the Google tools in some class or another whether it's Docs or Sheets. And so, I wasn't too worried about dealing with the technology. After all, I was giving them an already filled out sheet with the data. All they had to do was learn how to select columns of numbers and apply them in formulas which I had demonstrated in lecture earlier. I don't think it was a stretch for them to deal with the technology. At least my intention was to not make it about the tech, but to make it about the technique and the methods and the ideas and the concepts.

I: Right, I think that relates to one of the follow up questions here which was to what extent are those tools and software important for your teaching? I think you iterated that it's not about the tools, it's about learning the actual techniques.

UCSB05: For this class, but for example I've taught workshops for grad students on data acquisition from social media. And I've shown them how to do it, using an API to get tweets. And I showed them how to do that using R and I showed them how to do this using Python. Now, these were grad students in the social sciences, but they were pretty sophisticated when it came to knowledge of technology, even if they weren't programmers, they could deal with what I was telling them. This is a different audience, these are undergrads, and my purpose was different. The purpose of the grad student workshop was to introduce them to different tools and techniques. With the undergrads It was really more about the methods than the techniques.

I: Makes sense. Did you face any other challenges with their ability to work with data, rather than the tools?

UCSB05: Yeah, I would really like to do more with data visualization beyond just simple generation of bar charts and pie charts and scatter plots, which is really what a tool like Excel or Google Sheets that's really all they can do. And so, for the future classes I want to look into incorporating other tools for visualization like tableau, or something like it. Because I think data visualization can be a super interesting medium to teach them about big data actually because it speaks to communicating concepts, as well as dealing with tech, and it has this element of artistry to and having an eye for design and having an understanding how that works with human cognition, I mean it's a lovely field that that just incorporate so much. But in order to do that effectively, I really do need a better tool, a more sophisticated tool than Google Sheets. So, I'll be looking into something like Tableau or something else like that to help me. I'd love to hear what other people might have in terms of suggestions I don't know if you do.

I: I was working with, in my grad degree, working as a TA for a design lab and a lot of what we did was teaching students how to do data visualization. And even though it was a year ago, it feels like forever ago, but maybe we can talk about that another time.

UCSB05: Yeah, yeah, and I'd love to compare notes with you and if you have that class syllabus somewhere. I'd love to take a look at it.

I: Well, that wasn't a class it was more like a consultation lab for students to come in and get help on their homework and yeah, we did go and do instruction sessions and classes, too. And I'm not sure if I still have some of those slides but I'm sure they'd be happy to share them as well. Yeah, they've done some research on, especially the human cognition side but the design side too.

UCSB05: So, yeah, I found this really great textbook that I photocopy the chapter and have them read it it's by Andy Kirk on data visualization, and it really just encapsulates all these different interdisciplinary areas that play into dataviz and I know there are departments on campus that do more with that. And I've just emailed back and forth with a couple of professors, but I'd love to find out more what's happening there too. I forget the department's name. Their name, if it's MAT, it's a grad program in design and art.

I: And that's here? I guess I don't have a full familiarity with all the departments here yet.

UCSB05: Yeah, I've only been here four years myself. And there's a surprising amount of stuff. Yeah, it's [department redacted]. So it's [website redacted] that's their website check it out.

I: All right, I'll have to take a look at that. I've heard that here and there but I really haven't looked into that department yet. Kind of changing directions I guess I know you said that, like, [discipline redacted] is really quantitatively focused here.

UCSB05: The research.

I: The research okay. the next question is how do the ways in which you teach with data relate to those learning goals in the [discipline redacted] field.

UCSB05: So the learning goal, the stated learning goal for, say [course redacted] is to understand the different issues surrounding the use of big data in society today and more specifically income research, which is why we talk about things like ethics, like societal impact of data use and big data use, and gosh there's an endless amount of things we can talk about there in class. And among other things, the goal there is to have them have an idea of and an appreciation of an understanding of the technology that's involved and what it requires, and which is why a lot of what I talked about is big data and they've heard the term big data but what does it really mean? Like did you know that Twitter produces five petabytes of data a year, well what does that mean? How many high-definition movies is that right? I try to put that into an everyday perspective for them so they have a sense of the scale and really the enormity of this thing. And then realize that that data is all about you. That's what that is, all of that mountain, those mountains of data, anyway. So, the stated goal here, the goal of the class is to give them an awareness and an understanding of the high level topics around there. Be it ethnic societal impact, or technology and/or the computational skills required to work with this and navigate this and make sense of it.

I: Okay. Yeah, that makes sense. And this is moving on to a different topic here. But do you find that instructors in [discipline redacted] face ethical challenges in teaching with data? I guess more specifically with teaching with data, rather than using data in research but I think either is relevant.

UCSB05: Um, do I think the face, ethical?

I: Yeah, ethical challenges in teaching with data, like the data set that you're using or I guess even questions that come up with big data in that course.

UCSB05: Yeah. So, in this course, in [topic redacted], we certainly spend a lot of time discussing the ethics of data collection, data use. And, you know, more to the point data misuse. So that's really very explicitly talked about in this course. In other courses, I don't know that there are many if any [discipline redacted] courses that utilize big data sets with the undergrads. So, I'm not aware, actually, of any [discipline redacted] class that utilizes big data sets. But I do know that issues of the ethics of data use and ethics in research are very strongly underlined in the [discipline redacted] department. And in fact, in that research methods class I that I teach, it's a pre major course. [discipline redacted] has a pre major, which they have to take these four courses and have a B average before they're admitted into the major. Anyway, that is one such course that they have to do well in. We certainly talk about ethics in research and ethics and data use. So, I can attest to the fact that this is an area that is talked about to the students in more than one class. But actually, having students go and collect data, like big data sets, I don't know that any class does that.

I: Okay. Yeah. And have you seen any, you know policies or cultural changes here at UCSB with that or you know even wider [discipline redacted] field wise, that influence those ways in what you teach with data? Like there hasn't been any change over the years in that?

UCSB05: Well, just in the last year or so, we've started the Data Science Initiative on campus. And there's a push to turn that into a major eventually. So there's certainly been a cultural shift. When I joined the faculty four years ago, right after my PhD, I had a hard time frankly talking to people about what data science is and my PHD [field redacted], a lot of people were like "what is that?", they had no idea. Which was surprising coming from other places, you know, other universities have had information schools for, you know, for a while. Berkeley, Madison, Rutgers. Gosh, I mean I could, I could rattle off a dozen names. Yeah, so UCSB has come a long way just in the last two years at least with the awareness of the importance of data science in the world and not just in like stats and not just in computer science, but in the social sciences and the humanities. And so, you know, the faculty involved with this from all these different parts, they're very savvy about all the nuances of the use of data science in their own fields. So maybe that knowledge was always there but I think that the Data Science Initiative has kind of brought people together and brought it to the forefront. And computational social science has become more of a buzzword on campus now, at least in the past two years. This is just my feeling again. People are talking about that more in the social sciences. I've heard about more and more hirings in that area than before. But again, this is just my personal impression.

I: Yeah, and I think you know the majority of students we have coming to the R and Python carpentry workshops are from the social sciences here too.

UCSB05: Yeah, there's a real, you know, what's the word I'm looking for, demand for those courses from the social sciences, especially the grad students.

I: I mean, these online workshops we've been doing, especially registration will fill up within an hour of when we email them out. So, I know there's a lot of demand there. You know we always have students asking for more and of course there's only so many of us to teach right.

UCSB05: Well, if you guys ever need more instructors please reach out. I'd be happy to help out.

I: Okay. Sounds good.

UCSB05: Yeah, I mean one of the courses I teach [redacted], which is open to everyone on campus so anyone can take that and it's highly impacted. And that just goes to speak to what you just said, you know, everyone wants to learn, at least take an introduction to programming, even if they don't quite know what it means. And mind you, many of them once they learn they go "oh, that's not what I thought it was, I don't want to do that."

I: Oh they don't want to do that? I had the opposite happen in grad school.

UCSB05: They're like "What, it involves math? Logic? No, I don't want to do that"

I: That's funny. All right, I guess moving back to the courses that you teach. Do other people help or provide instruction with those like TAs or co-instructors, you know librarians, or do you send them to any kind of drop in sessions on campus?

UCSB05: I usually have a TA in any class that's over 60 students. If they have the budget for it, they will give us TAs. So, I did have a TA help me out in this class and you know their duties are really mostly grading and holding office hours. Again, if I was able to teach this class, you know, non-Covid, you know in person, I would love to have the TAs use their office hours as kind of like walk in labs, really they would actually demonstrate a lot of this stuff if they could. These grad students are our [discipline redacted] grad students who have an interest, a research interest in computational social science so they usually know their way around a lot of the software and have been super helpful.

I: Yeah, nice. Sixty students until you get a TA that sounds like a huge workload.

UCSB05: I know, that's another topic of discussion, that's a totally different time. UCSB has huge classes, and they're only getting bigger.

I: And I guess the next question here is, do you know of any other ways that students are learning to work with data or, you know, learning to program or code outside of their formal coursework in your class? Does it seem like people come in and, you know, either before or during utilize other sources to get a grasp on things.

UCSB05: Yeah. Again I don't have a big sample of students to, you know, talk about here, given this is a new course. But I have, and I'm sorry I maybe I misunderstood your question, but I did not see students go outside of class to learn more about especially the technical aspects.

I: Yeah, I'm thinking like Datacamp or coding camp, you know, those online resources.

UCSB05: Right. I've had a couple of students express interest but literally just two students interested in taking courses like that after. I've recommended, you know, paths that they could take, you know what courses to take. I don't believe they acted on those. Both of these students are graduating seniors and I am pretty sure they're busy with their own budding careers right now.

I: Do you often encourage people to do that kind of, you know, outside work outside of the courses? Especially if they're interested in coding or programming I guess that's where I'd see that naturally happen.

UCSB05: Absolutely, and then, I've had a couple of students take like directed research courses with me. Where you know we can set our own agenda. It's kind of like a, what's the word, yeah, directed research. Where we sit down and we talk about what it is that you want to do. Should we like, do a research project, do you want to learn more about this or that. And a couple of them wanted to learn more about programming. And I tailor the, you know, one on one intro to Python. And they did very well and they were very motivated and they understood it. So yeah, but that's a minority that's literally one or two students out of hundreds.

I: Yeah, okay. Yeah, I imagine they're really just too busy to think about other things like that and add them to their plates.

UCSB05: I mean it's a combination of they're too busy and they don't really know where to go. Yeah, but you know whenever they ask, I am happy to help. In fact, I actually put it in my syllabus that if they ever want to learn the skills beyond what this class teaches, I'd be happy to refer them. And, you know, there are a lot of online resources for that. Some of them are free or very inexpensive. A lot of the Khan Academy courses are, you know, good intro courses and there's a bunch of stuff on YouTube, and I so I do point them out to these things.

I: And then as far as your own training, I know it sounds like you haven't been here all that long and you had just finished your PhD before that. But have you done other training yourself since you did that PhD?

UCSB05: Training on what?

I: Training on you know, working with data or more training on programming or coding. I think I imagine not since you just recently completed your PhD, I think this question is more oriented towards people that are further along in their careers.

UCSB05: Yeah, and also because I, the nature of the classes that I teach, have kept my coding skills, pretty fresh. So, if this is working with data and coding with data is something that I do on a very very regular basis, both for my own research, which is whenever I have the time I engage with, and in the courses that I teach so that actually works out well for me.

I: Yeah, so your, your learning is built into the teaching essentially.

UCSB05: Yes, yes. Having said that, I always keep an eye open to new things that are coming out. I keep up with the literature. I always query my, especially my students who go out and find, you know who go out and work in industry. What companies like Google and Facebook and Microsoft are doing with data, what tools they're using. I try to stay, you know I try to self educate myself on what's going on there, on the trends and all of that.

I: Do you, it sounds like you do talk with other professors at UCSB about the courses that they're working on. But do you do a lot of sharing of syllabi and resources for instruction too?

UCSB05: Yes. I found the community here super helpful and willing to share ideas and syllabi and even slides and materials. So, yeah. And I do the same when someone asks, I share my stuff willingly yeah you know that it's part of the culture here.

I: Yeah, yeah. I'm seeing that, the only problem is how busy people are, right. It limits the time.

UCSB05: No everyone's crazy busy here and especially people who are immersed in research. And you know we're a premier research institute, I mean these people are busy and scarily productive, you know like, generating dozens of papers every year right that's unheard of at other universities.

I: Yeah, I can't imagine. Then the final question here is about, you know, considering those evolving trends. What types of training or assistance do you think would be beneficial to other instructors, when they're teaching with data?

UCSB05: I think teaching with data, especially teaching data analysis or data techniques, the training really is about doing the work yourself. Coding is a skill, like playing the piano, you can read books about it but you're not really going to learn how to play the piano unless you sit down and play it. Same with coding and working with big data and working with certain technologies, it's just stuff you have to do, whether it's in your own research, or if you just take a course or a couple of workshops. So, that's what I would recommend an instructor to have as background before they teach a course like this.

I: Yeah, I think the same could be said for stats. I feel like that's why it doesn't stick because you're often just being lectured about it but you don't actually try using the methods on things and that's really where you end up learning it.

UCSB05: So true, a lot of applied STEM is really about building skill, whether it's programming or using applied math or applied physics. There are theoretical aspects to all of those but the applied stuff is very much skill based. Just putting in the hours.

I: Yeah, that's interesting. Well, that's all I have for questions, is there anything that came up from the questions that you think you want to add in or anything from other experiences or perspectives that you might want to share?

UCSB05: No, I mean, I'm very glad I'm very excited that this activity is on the rise at UCSB. And I think the demand for classes like the workshops that you guys do in the library and courses like this data science course in the social sciences, that demand is only going to grow. And so, you know, this study that you and your colleagues are conducting is super timely. I'd love to know more about what your results are, what you find.

I: Yeah, we'll definitely send you the report when it's done both the local one. And the final one. It will take some time. I think the timeline is to have the local report done sometime I think late next spring.